

RISE — DECISIONS.md

The Single Source of Truth — Read This Before Any Part

STATUS: Living document. This supersedes conflicting statements in every numbered Part, including Part 12's own reconciliation table (which is carried forward here, verified against the actual files, and extended to cover Part 13 and everything decided since).

RULE FOR ANY CODING AGENT: If a Part contradicts this file, this file wins. If this file is silent, the Part stands.

SECTION 0 — Numbering Note

Part 12 is titled "PART 12 OF 12 — THE CAPSTONE."
Part 13 exists and
was written after it. The "OF 12" headers in Part 12
and earlier

parts are cosmetic errors — treat the project as 13 parts, soon to be restructured further as rewrites land (see Section 8).

SECTION 1 — Carried Forward From Part 12 (Verified Against Source)

Part 12 already resolved these conflicts correctly. Nothing below in this section is new — it's confirmed accurate and repeated here so this file is a complete reference on its own.

1.1 Subscription Model

Philosophy: the "becoming" layer is free forever. The "understanding" layer is paid.

- **FREE (Starter):** wake games, biometric lock, meal signal tracker
(see Section 4 — replaces "barcodeCalorieScanning"), last 14 days
sleep history, goals, habits, quotes, proof wall
- **CORE (\$2.49–\$5.99/mo, PPP-adjusted):** full
sleep history, weekly
AI review, sleep-nutrition insight

- **PRO (\$3.99–\$9.99/mo):** daily AI morning briefing, monthly letter, chronotype optimizer, sleep debt ledger, smart home integration
- **GUARDIAN (\$6.99–\$16.99/mo):** child profiles, parent dashboard, guardian notifications, + all Pro features
- Lifetime tier = permanent Guardian-level access

1.2 Referral Service

Use `referral_service_v2.dart` (Part 10A) only. Do not implement Part 6B's `referral_service.dart`.

1.3 Paywall Screen

Use `lib/features/paywall/paywall_screen.dart` (Part 10A) only. Part 6B's paywall inside `referral_screen.dart` is superseded.

1.4 Subscription Provider

Use Part 11's full implementation of `subscription_provider.dart`, not Part 2's stub.

1.5 Home Screen

Use Part 9's full `home_screen.dart`, not Part 1's stub.

1.6 Calorie Models — SUPERSEDED AGAIN, see Section 4

Part 12 resolved Part 2's stubs vs. Part 10B's full models in favor of Part 10B. That resolution is now itself superseded by the meal signal tracker decision below — do not build Part 10B's full calorie tracker as designed. See Section 4 for the current model.

1.7 Box Names

`AppConstants.sessionsBoxName =`
`'sleep_sessions_v1'` is canonical.

Any reference to `sleepSessionsBoxName` should be updated to this.

1.8 Bottom Navigation

5 tabs: **Alarms** | **Sleep** | **Goals** | **Nutrition** | **More**.

"More" opens a

bottom sheet (not a separate screen) holding Quotes, Leaderboard,

Wallet, Settings, Upgrade — confirmed against the actual wired

`main_shell.dart` in the Part 12 rewrite. That file's sheet was

missing a Referrals entry despite

`referral_service_v2.dart` being a fully decided, real feature with no navigation path to reach it — fixed in the Part 12 rewrite. "Profile" (mentioned in this section before it was verified against real code) doesn't have its own entry either — presumed folded into Settings, not confirmed as a bug.

1.9 Pubspec Versions

- `purchases_flutter: ^8.0.0` (not Part 10A's `^6.19.0`)
- `firebase_auth: ^5.2.1` (not Part 10A/11's `^4.20.0`)

1.10 Onboarding Route

Part 1's placeholder `/onboarding` route is replaced by Part 11's full `OnboardingShell`.

SECTION 2 — New Gap: Part 13 Was Never Reconciled

Part 13 shipped after Part 12's "complete" `pubspec.yaml` and file

tree were declared final. These need to be merged in:

New pubspec entries (add to Part 1/Part 12's master pubspec):

```
just_audio: ^0.9.40
audio_session: ^0.1.21
flutter_beep: ^1.0.0
```

New Hive boxes (add to the master box-opening sequence):

```
stakesBoxName      = 'alarm_stakes_v1'
charitiesBoxName   = 'charity_partners_v1'
chronotypeBoxName = 'chronotype_v1'
```

Tier-flag gap: Part 13's stakes feature

(stakeMaxGuardian = 5000

cents, i.e. tier-dependent stake caps) implies stakes should be a

tracked capability per tier, but no stakesEnabled flag exists in

Section 1.1's tier model. Decision: stakes stays

disabled at the

code level for all tiers until the legal audit (Section 5)

clears

it — see Section 5 for what that means concretely.

When cleared, add

stakesEnabled: false (all tiers) to the flag model now, flip later.

typeld assignment — CORRECTED (this note was wrong in an earlier

version of this document): the real breakdown, confirmed by reading

Part 13's actual code rather than its header summary, is **five**

Hive types across three files, not three: 16 =

`StakeStatus` (enum),

17 = `AlarmStake`, 18 = `CharityPartner`, 19 =

`ChronotypeType` (enum),

20 = `Chronotype`. There is no `ChronotypeProfile` class — that name

was invented in an earlier pass and has been corrected everywhere it

appeared (Part 12 rewrite's adapter list and file tree).

`chronotypeBoxName` **is consistent, not a conflict** — it's the

Part 13 implementation of the

`chronotypeOptimizer: true` Pro flag

already defined in Part 12. No action needed beyond noting it.

SECTION 3 — Architectural Pivots (Decided This Conversation)

These are additive/replacing layers on top of the existing services,

not a teardown of what's built. Each gets its own Part when we get to rewriting; this section is the decision record, not the full spec.

3.1 Agentic Core (biggest single change)

New service: `ai_planner_service.dart`. Reads sleep history + calendar

- meal signals + habit data → proposes alarm time/mode adjustments → user approves or overrides → planner logs the override and adapts.

This sits alongside (does not replace) the existing `ai_reflection_service.dart`, which continues to handle the generative text features (briefing copy, letter copy).

3.2 Wearable-First Data Architecture

New abstraction: `WearableDataBridge` — Apple Health (HKCategoryType .sleepAnalysis), Google Health Connect (SleepSessionRecord), Oura API, Fitbit Web API. When wearable data is present, it is the primary sleep data source; phone accelerometer (`motion_analyzer.dart`, Part 5)

becomes the fallback, not the foundation. UI labeling changes accordingly: "estimated from movement" (no wearable) vs. "from your [device]" (wearable present) — never claim clinical sleep staging from accelerometer alone.

3.3 Voice Layer

New service: `voice_interaction_service.dart`. On-device speech recognition (`SFSpeechRecognizer` iOS / `SpeechRecognizer` Android, no internet required) for: wake-word alarm dismissal confirmation, a voice-based wake protocol as an alternative to the typed math game, and spoken delivery (TTS) of the morning briefing. Replaces the `url_launcher` Alexa/Google Home deep-links with actual on-device handling where possible; Alexa/Google Home deep-links remain as a fallback for users who prefer their existing smart speaker setup.

3.4 Matter-Based Smart Home — CORRECTED, see Part 12 rewrite

This section was wrong. It was based on the original critical-

analysis document's claim that smart home was a non-functional

`url_launcher` placeholder. Having since read

`smart_home_service.dart`

directly (Part 12 rewrite): it's real, working code —

Philips Hue

local REST API with bridge auto-discovery, LIFX cloud

API, SwitchBot

kettle control, WiFi-gated so it never fires on mobile data, with

genuine call sites in AlarmService (light ramp before alarm) and

WakeConfirmationService (full brightness on wake).

Nothing here needs

fixing.

Corrected plan: Matter becomes a **fourth, additive**

`SmartHomeProvider` option (universal compatibility)

alongside the

existing three, not a replacement. Lower priority than originally

framed — the feature already works without it.

3.5 On-Device AI Split

New service: `local_ai_service.dart` routes

prompts by sensitivity and

complexity:

- **On-device (free, private, all tiers):** dream journal tagging, habit check-in prompts, personalized wake-game question generation
- **Cloud (Pro-gated, existing model):** weekly review, monthly letter, sleep-nutrition correlation analysis

Note the deliberate exception: the **Monthly/Weekly Letter stays**

cloud-generated, not on-device. Current on-device models (Apple Intelligence / Gemini Nano as of this decision) are good at classification and short prompts, not yet at the quality bar the Letter needs to land emotionally. Revisit this if on-device model quality changes materially.

3.6 Biometric Identity – Primary/Secondary Swap

`local_auth` (Face ID / Android biometric) becomes the **primary** alarm-dismissal identity layer. The PPG camera system (Part 4) becomes an **opt-in Premium novelty layer** ("prove it's really you with your heartbeat"), not the reliability foundation.

PIN remains
the fallback for both. This changes framing and
default flow in
`biometric_service.dart`, not the underlying PPG
engine itself –
that code is still valuable, just repositioned.

SECTION 4 – Calorie Tracker → Meal Signal Tracker

**Decision: Part 10B's full calorie tracker (barcode
scanning, Open**

**Food Facts integration, photo food recognition) is not
built as
originally specified.** Replaced with:

Meal Signal Tracker – 5 inputs, ~20 seconds to log,
free tier:

1. Last meal time → time picker
2. Meal size → Light / Medium / Heavy
3. Alcohol units → 0 / 1 / 2 / 3+
4. Caffeine after 2pm → Yes / No
5. Water today → Under / On target / Above target

This feeds the sleep-nutrition correlation engine (Core
tier feature,

Section 1.1). New model replaces

`calorie_entry.dart` / `food_item.dart` :

`meal_signal_entry.dart` — same type of slots (3, 4)

can be reused

since those are being freed up by dropping the full tracker.

What's dropped entirely: building a competing food database,
in-house photo food recognition.

What's kept as optional, not core: a partner API integration
(Passio or Calorie Mama) for users who specifically want photo food logging, offered as a Pro add-on rather than a built-in system —
build vs. partner decision, not a free-tier expectation.

What's still worth building, later, as a differentiator:
the
2,000-item curated regional food database
(West/East African,
South/Southeast Asian, Middle Eastern staples) for
users who want
richer logging than the 5-signal tracker but aren't
served by MFP's
crowd-sourced entries. This is a Section 8 (later) item,
not urgent.

Tier flag update: `barcodeCalorieScanning: true`
(free tier,
Section 1.1) is renamed/replaced by
`mealSignalTrackerEnabled: true`
(free tier). Barcode scanning is no longer a flag at all
unless the
partner API add-on above gets built.

SECTION 5 – Legal & Compliance **(full detail in** **RISE_COMPLIANCE_AUDIT.md)**

Architecture decisions that follow from the audit,
restated here so
they're part of the source of truth:

- Face ID/biometric OS auth is primary identity
(Section 3.6) –
reduces GDPR Art. 9 exposure vs. PPG-as-primary
- PPG enrollment data: on-device only by default;
cloud sync requires
a separate, explicit consent screen distinct from
general ToS
- Biometric right-to-erasure flow is a Part 4
requirement, not a
future add-on

- Alarm Stakes (Part 13): **disabled at code level** (Section 2) until gambling/money-transmitter/charitable-solicitation legal opinions come back. Age-gating (18+) and self-exclusion scaffolding should still be built now, since they're needed regardless of the eventual legal structure.
 - `MedicalDisclaimer` component required at three touchpoints: biometric enrollment, sleep score display, heart rate display
-

SECTION 6 — Admin Dashboard (new system, not in original 13 parts)

New internal tool, RBAC-tiered:

- **Tier 1 (support):** user search, ticket tools, common fixes (resync alarm, resend receipt, extend trial) — scoped to ticket-relevant data, no raw biometric/health data by default
- **Tier 2 (senior support):** refunds, subscription overrides,

stakes dispute resolution (once stakes is live)

- **Tier 3 (admin):** feature flags, pricing config, biometric/health data access (logged, justified per access)
 - All tiers: every action immutably logged (who/what/when); destructive actions require confirm step
-

SECTION 7 – Monetization Additions

On top of the existing tier model (Section 1.1):

- Weekly Letter precision paywall: free tier gets it monthly, Core/Pro gets it weekly (uses the existing monthlyAiLetter-style flag pattern)
- Family/couples plan: new pricing tier variant, higher ARPU per household, ties to the (already-planned) couples sleep score feature
- B2B2C employer/insurer wellness licensing: explored as a parallel sales channel, not a code change
- Affiliate commissions (wearables, sleep gear): allowed, but must be clearly disclosed and structurally firewalled from the correlation

insight engine – the AI's recommendations can never be influenced by affiliate relationships. This is a hard rule, not a guideline.

- Anonymized aggregate data licensing: allowed only with explicit opt-in, true aggregation (no re-identification path), and public disclosure – treat as high-reputational-risk, low-priority

Marketing rule: no competitor trademarks anywhere in App

Store/Play Store metadata (Apple's Guideline 2.3.7 explicitly bars this and it risks both rejection and legal exposure).

Generic category

keywords are fine and encouraged: "calorie tracker," "smart alarm,"

"sleep score," "wake up light," "meal timing."

Superiority claims are

made only for what RISE actually and verifiably does differently.

SECTION 8 – What's Still Open (not decided yet)

- Exact on-device model choice (Gemini Nano vs. Phi-3 Mini fallback tier) — needs a device-coverage tradeoff analysis
- Exact Matter library/plugin for Flutter — `flutter_matter` (federated: `flutter_matter_android` + `flutter_matter_ios`) confirmed to exist as of the Part 1 rewrite; exact current version still needs verifying at implementation time
- Jurisdiction-specific legal opinions (biometric, stakes) — blocking, not yet started
- Cultural regional food database — later-stage, not urgent
- Whether to renumber Parts 1–13 into a cleaner structure once rewrites land, or keep sequential numbering with corrections in place. Default until decided: keep sequential numbering.
- `google_mobile_ads` fate — its only use case (ad-unlock for the old full calorie tracker) is gone now that the meal signal tracker (Section 4) has no natural ad-unlock moment. Dependency and Android manifest entry both kept in place but unwired

pending a real
decision — found during the Part 1 rewrite.

- `wearable_data_bridge.dart`'s sample-aggregation logic (merging overlapping HealthKit/Health Connect intervals into one nightly summary) is specified at the interface level only — needs full implementation against the real HealthDataPoint shape once the `health` package version is pinned — found during the Part 5 rewrite.

SECTION 9 — Rewrite Sequencing

1.  **Part 1** — DONE. See `RISE_PART_1_BLUEPRINT_REWRITTEN.md`.
2.  **Part 4** — DONE. See `RISE_PART_4_BLUEPRINT_REWRITTEN.md`.
Delivered as a changeset (only changed files reproduced in full; unchanged files explicitly listed rather than re-copied).
3.  **Part 5** — DONE. See `RISE_PART_5_BLUEPRINT_REWRITTEN.md`.
Changeset style, same as Part 4.

4.  **Part 10B** – DONE. See
RISE_PART_10B_BLUEPRINT_REWRITTEN.md.
Meal signal tracker replaces the calorie tracker;
food_item.dart
kept dormant (not deleted) for the later cultural
food database;
ai_reflection_service.dart updated (stale model
string, meal-signal
integration, softened correlation-insight prompt).
Found and fixed
a real box-name bug (calorie_service.dart
referenced box names
Part 1 never declared).

Part 8 – CHECKED, no changes needed.
"Barcode" there is the
wake-up game mechanic, unrelated to food
logging; "financial
stakes" has no live implementation in that file.

5.  **Part 12** – DONE. See
RISE_PART_12_BLUEPRINT_REWRITTEN.md.
Bigger job than originally framed – turned out to
be the true
consolidation point for pubspec.yaml, the Hive
adapter registry,
and box names (Part 1 was never meant to be
complete on any of
these, same pattern as home_screen.dart). Found
and corrected the
smart-home false premise (Section 3.4), the

missing Referrals nav
entry (Section 1.8), and locked in Part 13's typelds
(Section 2).
RECONCILIATION.md now points to this
document instead of
duplicating it, closing off the exact failure mode
that let Part 13
drift out of sync originally.

6.  **Part 13** — DONE. See
RISE_PART_13_BLUEPRINT_REWRITTEN.md.
Client-side legal gating is now code-complete: a
hard compile-time
kill switch (`_stakesCompiledGate = false`), age
gate, self-
exclusion, and a dispute-filing path that routes to
human review
instead of resolving anything automatically. Still
fully blocked on
the actual legal opinions (Section 5) before that
constant flips —
this rewrite makes the feature safe to leave
dormant in the
codebase, not legally cleared to launch. Also fixed
the typeld bug
this document previously had wrong (see Section
2), and wired up an
"Add a stake" entry point in Part 8 that the original
Part 13
always assumed existed but was never built.

Open, outside this project's reach: the Cloud Functions (`initiateAlarmStake.js` etc.) are server-side and need their own independent enforcement of the same flag — a gated Flutter client doesn't stop a direct API call to those functions.

SECTION 9 IS NOW COMPLETE FOR ALL 13 NUMBERED PARTS

What's left has no existing Part to rewrite — it's new work:

7. **New parts** — agentic core (3.1), voice layer (3.3), smart home (3.4), on-device AI (3.5), admin dashboard (Section 6)

Format note: changeset style (full content only for files that actually changed, explicit "no change" notes for the rest) proved faster and clearer than Part 1's full reproduction across Parts 4, 5, 10B, 12, and 13 — recommend the same approach for the new (unnumbered) parts ahead, even though those are greenfield rather than edits to existing files, since the same "state what's new vs.

what it plugs
into unchanged" clarity still applies.